Funding intent, made explicit

Private capstone research | 7 November 2026 | Priorities 2 and 5. Defaults are unchanged.

The new optional rule screens whether the fixed cash-first trading policy makes EVERY dated payment in full on at least 99% of modeled paths. It is separate from the existing tail-shortfall allowance, which can pass even when payments are missed. This is a finite candidate screen, not a new globally optimal chance-constrained solver.

Three distinct verdicts

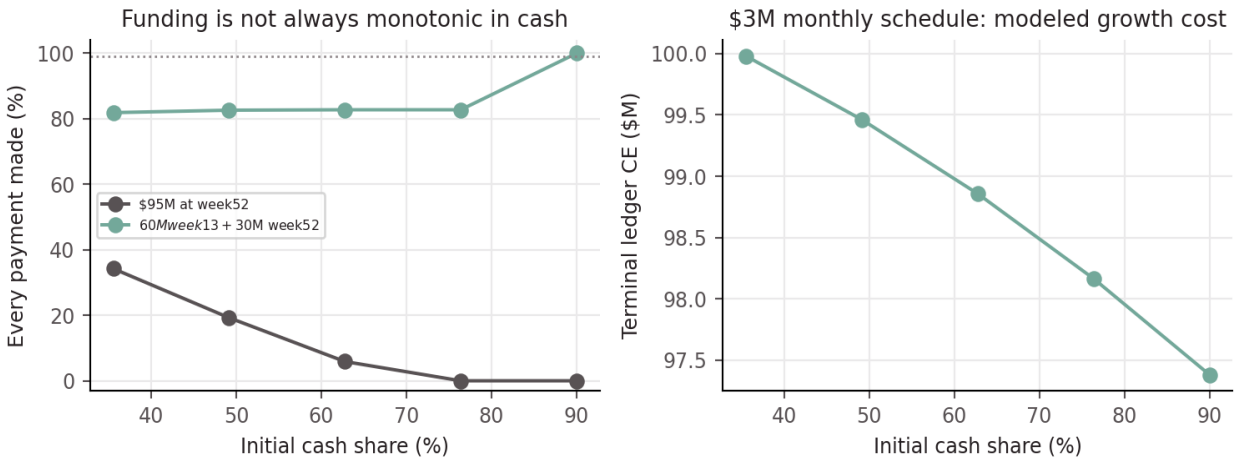
Check	Meaning
Tail budget	The declared size of the bad-tail funding gap stays inside each budget. Can permit unpaid calls.
Strict model screen	Joint paid-all event; reference one-sided 95% binomial lower bound >=99%, plus minimum nominal/stressed weighted batch rates >=99%.
Final all-risk verdict	Separate frozen final tests also require dated funding, path-loss, terminal-loss and holdings limits. A strict funding pass alone is insufficient.

Current four-batch evidence

Illustrative schedule	Strict selection / final outcome
$3M:12 monthly calls	Raw candidate selected;1600/1600 final reference paths fully paid, lower bound 99.8129%. Separate final path-loss check FAILED. Not a safe recommendation.
$95M at week 52	No validation candidate.90% cash +10% unsellable property misses $5M on every path. More cash cannot manufacture the missing liquid capital.
$60M at week 13 +$30M week 52	90% cash +10% property anchor is first strict candidate;1600/1600 final paths fully paid, lower bound 99.8129%; final all-risk checks passed under this fitted law.

All amounts are USD. These are counterfactuals, not user obligations. Current default law remains the iid Gaussian scale mixture. Final paths were not used to choose or repair portfolios. No economic 99% guarantee, no W 1 guarantee for the cash ledger.

Liquidity versus modeled growth



Each plotted point is the same frozen validation candidate/anchor grid:100%,75%,50%,25%,0% candidate, with the rest in cash plus required fixed holdings. Reference paid-all rates and uniform-reference cash-ledger CE use 800 validation paths. They are not final performance estimates or confidence bands.

The $95M fixture exposes the error in a simple "more cash is always safer" claim. Here risky growth is needed to exceed the $90M liquid-capital ceiling. Validation paid-all rates fall 34.25%,19.25%,5.875%,0%,0% as cash rises 35.54% to 90%. No candidate reaches the 99% screen.

For $60M early+$30M final, the four non-anchor candidates pay all on 81.875%-82.75% of validation paths. The 90% cash anchor reaches 100%. For the much smaller $3M monthly schedule, every grid point pays all in validation; modeled ledger CE falls from $99.98M to $97.38M as cash rises. Those CE values are not the optimizer objective and are not expected returns.

What the confidence calculation does

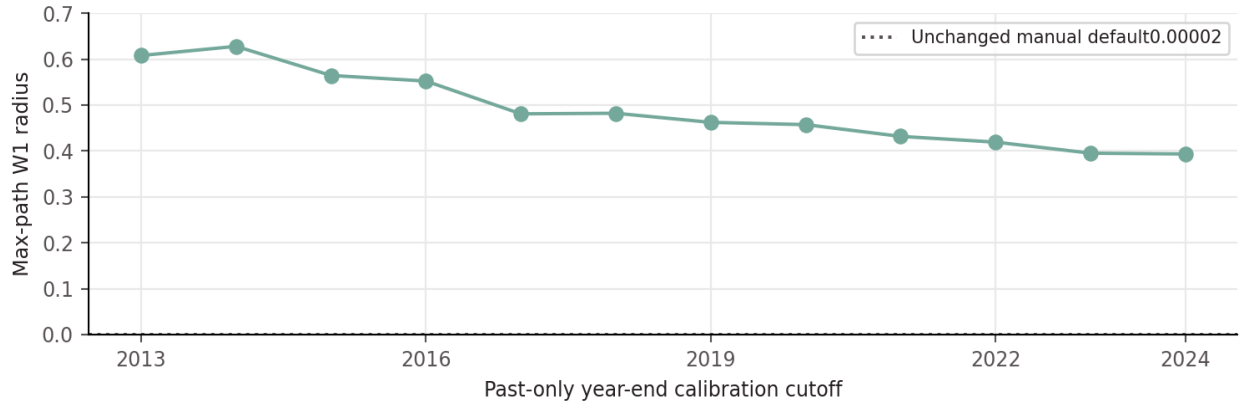
A path is one trial, with success only if EVERY dated payment is made, within $0.000001 roundoff. Assets and payment dates are not extra trials. With zero misses,299 independent paths is the minimum for a one-sided 95% Clopper-Pearson lower bound reaching 99%. The 600 path coarse final run has lower bound 99.502%; the 1600 path run has 99.813%.

That bound assumes independent generated paths conditional on frozen fitted inputs and the validation-selected fixed portfolio. It does not validate the economic law. The validation version is a selection screen, not post-selection coverage. EP-weighted nominal/stressed fractions are descriptive model rates, not binomial confidence bounds. Imported scenario archives are refused in this strict mode because their independence is unverified.

Past-only radius: the conflict is real

The optional empirical policy uses nonoverlapping 52 week actual-history blocks,64 fixed-seed balanced half-splits and exact uniform empirical max-path transportation distances. It chooses their 90 th percentile. This is a declared distribution-shift STRESS policy, not a 90% confidence set for the true future law.

Current observation	Result
History through 2026-10-30	21 complete annual blocks
Empirical radius	0.406744 vs unchanged manual 0.00002
Unavoidable funding buffer	$813.49M on $100M wealth
Smallest monthly-payment budget	$50,000
Verdict	Incompatible settings. No candidate and NO radius truncation.



2006-2012 year-end snapshots have fewer than 8 complete blocks and remain unavailable, with no zero/manual fallback. 2013-2024 snapshots are available but every radius conflicts with the current budget. Property uses latest-vintage quarterly URA data with an assumed 45 day release lag, not revision-safe vintages. Deterministic cash and liabilities show no observed displacement; that is not evidence of no uncertainty.

The current unrestricted-support W 1/CVaR envelope is too conservative at these observed historical distances for the existing budgets. A usable economic confidence policy would need new support/metric assumptions and stronger calibration evidence. Shrinking the radius until the allocation passes would hide the conflict.

Implementation and evidence

Added, without replacing

Two explicit controls: funding_mode=pay_all_99 and radius_calibration_mode=past_empirical. Original defaults remain tail_allowance and manual 0.00002. The fixed 10% property holding, no sale, zero rent and original risk budgets remain. The bounded annual-volatility alternative remains opt-in; optimizer/executable-ledger integration is still parked.

The UI displays strict and all-risk verdicts separately, the fixed-grid liquidity/CE tradeoff, weighted-rate caveats and empirical-radius conflicts. Mobile uses stacked candidate cards. Static export and live compute share the same data. Selection weights/configuration are saved before independent final paths are generated.

Checks and limitations

349 tests pass with 3 recorded warnings. Tests include exact binomial inversion/all-success formula, joint-date event, weighted stress rejection, insufficient sample size, archive refusal, the $95M unsellable-property failure, unchanged defaults, transportation assignment oracle, duplicate-sleeve/repeated-time invariance and future-cutoff exclusion. Measured corrected evidence batch: 3.39 seconds. The standalone wheel runs outside the source checkout; desktop/320 px layouts and live controls were checked.

One initial experiment setup error was caught: recurring-payment expansion replaced the named counterfactual schedules. Those initial fixtures are marked invalid and preserved. Corrected fixtures clear the recurring schedule and were rerun before reporting these numbers. A final print-only JSON formatting failure and initial mobile table overflow were also fixed. No failed final risk check was rerun away.

Method sources

Exact binomial intervals: R Clopper-Pearson reference

Binomial interval formulas: Minitab methods

Wasserstein chance-constraint methods and distance to unsafe sets

Wasserstein distributionally robust optimization: Esfahani/Kuhn

The papers do not certify this empirical half-split radius or transfer a CVaR envelope to the nonlinear cash ledger. Raw observations, fitted-law probabilities and economic guarantees remain distinct.

Private source of record: github.com/TijsdeBoer/emfin_capstone. No client data, public sharing, spending, contact or upstream changes.